

WORKPACK 3

Power Query: Clean It Properly

Level 3 Data Technician · GreenWheels Urban Mobility · Self-Diagnostic Pack

Get Ready to Smash This

A head-start check — this is not a test

Below are eight things this pack covers. Tick the ones you can already do.

If you can tick all eight: skip to the Mini-Mission on page 8 and prove it in two minutes. Then move on to Pack 4.

- ☐ I know where to find Power Query in Excel (the Data tab, Get & Transform Data section).
- ☐ I can load a table from an Excel sheet into the Power Query Editor.
- ☐ I know what the Applied Steps panel shows and why it matters.
- ☐ I can check and change the data type of a column in Power Query.
- ☐ I can filter a column in Power Query to remove or isolate rows.
- ☐ I can replace a specific value in a column using Replace Values.
- ☐ I can add a custom column in Power Query to flag records.
- ☐ I can remove duplicate rows in Power Query by specifying which columns define uniqueness.
- ☐ I can handle null (blank) values in Power Query — replacing, removing, or flagging them.
- ☐ I can close the Power Query Editor and load the cleaned table back into Excel.

Marcus from the next desk says: "Power Query is just for big data. For 40 rows you can fix things by hand." Power Query is also for having a documented, repeatable, reversible audit trail. Which Brian from Finance will ask to see.

What This Pack Is About

In Pack 2 you found six anomalies in the GreenWheels dataset by sorting and filtering. In this pack you will use Power Query to clean that data in a structured, documented way.

Power Query is a tool built into Excel. It records every change you make as a numbered step. This step list is your audit trail — proof of exactly what you changed and in what order. You can undo any step, re-run the whole process on new data, or show someone else exactly what you did.

By the end of this pack, you will be able to open the Power Query Editor, check data types, fix an incorrect value, remove a duplicate, handle null values, filter out invalid rows, add a flag column, and load the clean result back into Excel. The pack takes around 40 to 50 minutes.

The Connection to Pack 2

Pack 2 identified the anomalies. This pack fixes two of them, flags two more, and leaves two documented for investigation — because you cannot always correct data without knowing the right answer. The anomaly log you built in Pack 2 tells you which is which.

Fix in Power Query: HC-030 Zone value (entry error: 'West' should be 'South'). Remove HC-041 duplicate (Victoria Park Q3 2024 appears twice).

Add a flag column: HC-015 (hires = 1,847), HC-010 (revenue = £0), HC-019 (Active Bikes = 0 with hires), HC-031 (8 hires), HC-035 (duration = 0). Handle HC-042 null values.

Leave for investigation: The correct values for HC-015, HC-031, and HC-019 are unknown. Flag them but do not guess.

The Skills

1 Opening the Power Query Editor

A new window inside Excel

SCENARIO

You have the GreenWheels file open and your anomaly log ready. It is time to start cleaning. Power Query opens in its own window. The original data in Excel is not changed until you choose to load the result back.

Power Query lives on the Data tab. In the Get & Transform Data section, you will see a button called Get Data. For data that is already in your Excel file, the quickest route is: click anywhere inside the Hire_Data table, then go to Data → Get & Transform Data → From Table/Range. The Power Query Editor opens in a new window.

WORD TO KNOW Power Query

A built-in Excel tool for connecting to, transforming, and cleaning data. It records every action you take as a named step, so you always have a full record of what was done to the data.

WORD TO KNOW Power Query Editor

The separate window where you work on your data in Power Query. It has three main areas: the Queries pane on the left, the data Preview in the middle, and the Applied Steps panel on the right.

WORD TO KNOW Get & Transform Data

The section of the Data tab in Excel that contains Power Query tools. 'Get Data' brings in data from external sources. 'From Table/Range' loads an existing Excel table into the Power Query Editor.

TRY IT

Open urban_cycle_hire_aggregated.xlsx. Click any cell inside the Hire_Data table. Go to the Data tab. In the Get & Transform Data section, click From Table/Range. The Power Query Editor window opens.

Look at the three areas: the Queries pane (left), the data Preview (centre), and the Query Settings panel (right). Under Query Settings you will see Applied Steps — this is where Power Query records every change.

CHECK YOURSELF

What does the Applied Steps panel show when you first open a table in the Power Query Editor?

Answer: It shows the automatic steps Power Query added when it loaded the table — typically 'Source' (connecting to the data) and 'Changed Type' (Power Query's first guess at the data types). These are added automatically before you do anything.

2**Checking Data Types***Numbers must be numbers, not text***SCENARIO**

Before you clean any values, you need to make sure each column is the right type. A column stored as text cannot be summed. A date stored as text will not sort in date order.

Power Query guesses the types when it loads the data — but it sometimes guesses wrong.

Every column in Power Query has a data type shown as an icon to the left of the column name. A number icon means numeric. ABC means text. A calendar icon means date. Click the icon to change the type.

WORD TO KNOW**Data type**

The category of values a column is allowed to hold. Common types: Whole Number, Decimal Number, Text, Date. The type controls what you can do with the column — you cannot add text values together, for example.

WORD TO KNOW**Type icon**

The small symbol to the left of a column name in Power Query. It shows the current data type: 1.2 for decimal, 123 for whole number, ABC for text, a calendar for date. Click it to change the type.

For the GreenWheels data, the correct types are:

Text: Hire Record ID, Station Name, Zone, Period.

Date: Period Start Date.

Whole Number: Total Hires, Avg Hire Duration (Mins), Active Bikes.

Decimal Number: Total Revenue (£), Customer Satisfaction Score.

TRY IT

In the Power Query Editor, look at the type icon for each column. Check whether each matches the list above. If a column has the wrong type, click its type icon and select the correct one from the list.

If Power Query asks 'Replace current conversion step or add a new step?' — choose Add new step. This preserves the original automatic step and adds your correction on top.

CHECK YOURSELF

Why does it matter whether a numeric column is typed as Text in Power Query?

Answer: A column typed as Text cannot be used in calculations. SUM, AVERAGE, and other formulas will fail or return an error if the column contains text values rather than numbers. Fixing the type in Power Query means the correct type is applied every time the query is refreshed.

3**Replacing a Value***Fixing the zone entry error***SCENARIO**

Pack 2 found that HC-030 (South Station, Q2 2024) has Zone = 'West'. Every other South Station record has Zone = 'South'.

This is a clear entry error. You know the correct value. Power Query's Replace Values tool can fix it in one step.

Replace Values finds every cell in a column that matches a value you specify and replaces it with a value you choose. Unlike editing a cell directly, this action is recorded as a named step in the Applied Steps panel.

WORD TO KNOW**Replace Values**

A Power Query action that finds all cells in a selected column that match a specific value and replaces them with a new value. The action is logged as a step in Applied Steps.

To replace a value: click the column header to select the Zone column. Then go to the Home tab in the Power Query Editor → Replace Values. In the Value to Find box type West. In the Replace With box type South. Click OK.

IMPORTANT NOTE

Replace Values changes every cell in the column that matches — not just the one you are thinking of. Before you run it, check whether any other station legitimately uses Zone = 'West'. In this dataset, Westfield Hub, Canal Quarter, and North Bridge all have Zone = 'West', so this replacement would be wrong if applied to the whole Zone column.

Instead: filter the Zone column to show only the South Station rows first, then apply Replace Values — or edit the single cell directly in the source data and refresh the query.

TRY IT

In the Power Query Editor, select the Zone column. Go to Home → Replace Values. In 'Value to Find' enter West. In 'Replace With' enter South. Click OK. Look at the Applied Steps panel — a new step called 'Replaced Value' has been added.

Check the Preview: does the Zone column still contain any 'West' values that should remain? (Westfield Hub, Canal Quarter, and North Bridge should still show 'West'.) If they have been changed incorrectly, click the X next to the Replaced Value step to undo it and use the direct cell edit approach instead.

CHECK YOURSELF

After using Replace Values on the Zone column, how can you confirm the change was recorded?

Answer: Look at the Applied Steps panel on the right side of the Power Query Editor. You should see a new step labelled 'Replaced Value'. Clicking that step shows you the data at exactly that point in the process — before any later steps were applied.

4**Filtering Rows in Power Query***Removing or isolating problem records***SCENARIO**

HC-031 (South Station, Q3 2024) has Total Hires = 8. Without knowing the correct value, you cannot fix it.

You can filter it out of the analysis dataset — and document in your anomaly log that you did so, and why.

Filtering in Power Query works like AutoFilter in Excel — but here it is recorded as a step. Every filtered-out row disappears from the preview and from the loaded result. The original source data is not touched.

WORD TO KNOW**Filter (Power Query)**

An Applied Step that removes rows from the query result based on a condition you set. The source data is unchanged — only the query's output is affected. The filter is recorded and can be removed by deleting its step.

To filter out rows where Total Hires is below a threshold: click the filter arrow on the Total Hires column header in the Preview. Choose Number Filters → Greater Than. Type a value such as 50. Click OK. Rows with 8 hires (HC-031) and potentially 45 hires (HC-034) would be excluded.

IMPORTANT NOTE

Before filtering out any rows, record your decision in the Anomaly_Log. Write which records were removed, which column triggered the filter, the threshold value used, and why. Removing rows without documentation is not validation — it is just deletion.

TRY IT

In the Power Query Editor, click the filter arrow on the Total Hires column. Choose Number Filters → Greater Than. Enter 50. Click OK. How many rows remain in the preview?

Look at the Applied Steps panel. What is the new step called? Now click the X next to it to remove the filter — you will add the correct threshold after the team has reviewed HC-034.

CHECK YOURSELF

What is the difference between filtering rows out in Power Query and deleting them in the Excel sheet?

Answer: Filtering in Power Query is non-destructive. The source data is untouched. You can remove the filter step at any time. Deleting rows directly in Excel is permanent — there is no automatic undo, and no record of what was removed or why.

5**Adding a Flag Column***Marking records for investigation***SCENARIO**

HC-015 (Market Street, Q3 2024) has 1,847 Total Hires. You cannot correct this value without knowing the right number.

Rather than removing the record, you want to keep it in the dataset but mark it clearly so anyone using the data knows it needs checking.

A custom column is a new column you add in Power Query, calculated from a formula or a fixed value. You can use it to add a flag — a simple label that marks certain records for attention.

WORD TO KNOW**Custom column**

A new column added in Power Query using the Add Column → Custom Column option. The column can contain a formula, a fixed text value, or a conditional result. It is added as a step in Applied Steps.

WORD TO KNOW**Conditional column**

A column added in Power Query using Add Column → Conditional Column. It assigns one value if a condition is true and another if it is false — similar to an IF formula in Excel, but defined with dropdowns rather than typed syntax.

To add a flag for records where Total Hires exceeds 800: go to Add Column → Conditional Column. Set the condition: if Total Hires is greater than 800, output the text 'Review: hires above valid range'. Otherwise output blank. Name the column 'Data Quality Flag'.

TRY IT

In the Power Query Editor, go to the Add Column tab. Click Conditional Column. Name the column Data Quality Flag. Set the condition: if Total Hires is greater than 800 then 'Review: hires above valid range', otherwise blank. Click OK.

Look at the Preview. Only HC-015 should show a flag in the new column. All other rows should show blank or null. Check the Applied Steps panel — a new step has been added.

CHECK YOURSELF

Why add a flag column rather than simply removing HC-015 from the dataset?

Answer: Removing HC-015 would make it disappear from all calculations, which could understate total hires for the year. A flag column keeps the record visible but marks it so analysts know to treat it with caution. The anomaly log records what the flag means and why it was added.

6**Applied Steps and Close & Load***Your audit trail, then back to Excel*

SCENARIO

Brian from Finance has asked to see exactly what changes were made to the data before the analysis was run.

The Applied Steps panel is your answer. It shows every action in order, named and reversible.

The Applied Steps panel shows the full history of your query. Each step has a name and can be clicked to see the data at that point in the process. Steps can be renamed to make the list more readable. Here is what a completed clean-up might look like:

1	Source — connected to the Hire_Data table
2	Changed Type — set correct data types for all columns
3	Replaced Value — Zone: 'West' corrected to 'South' for HC-030
4	Removed Duplicates — kept first Victoria Park Q3 2024 row (HC-007), removed HC-041
5	Replaced Value 2 — null values in HC-042 replaced with 'Not captured'
6	Added Conditional Column — Data Quality Flag for hires > 800
7	Filtered Rows — removed records with Total Hires < 10 (HC-031)

WORD TO KNOW**Applied Steps**

The numbered list of every action taken in the Power Query Editor, shown in the Query Settings panel on the right. Each step can be clicked to review, renamed for clarity, or deleted to undo the action.

To rename a step: right-click its name in the Applied Steps panel and choose Rename. Use plain-English names — 'Fix South Station Zone' is more useful than 'Replaced Value'.

When you are satisfied with the query, click Close & Load in the Home tab. Power Query closes and loads the cleaned data into a new sheet in Excel. The original Hire_Data sheet is not changed.

WORD TO KNOW**Close & Load**

The button in the Power Query Editor that applies all steps and writes the result to an Excel sheet. The original data source is unchanged. You can re-open the query at any time to edit or add steps.

TRY IT

In the Applied Steps panel, right-click each step and rename it to a plain-English description of what it does. Then click Close & Load (Home tab). A new sheet appears in your workbook containing the cleaned data.

Compare the new sheet with the original Hire_Data sheet. The Zone value in HC-030 should now read 'South'. The Data Quality Flag column should appear. The original Hire_Data sheet should be unchanged.

CHECK YOURSELF

If Brian asks to see the audit trail of your data cleaning, what do you show him?

Answer: Open the Power Query Editor (Data tab → Queries & Connections, then double-click the query). Show him the Applied Steps panel. Each step is named, numbered, and clickable — clicking a step shows the data at exactly that point in the process.

7

Removing Duplicates and Handling Nulls*The last two anomalies***SCENARIO**

Pack 2 found a duplicate record (HC-041 is a second Q3 2024 row for Victoria Park) and a record with four blank fields (HC-042, Meadow Park).

Power Query has specific tools for both — and both actions are recorded as named steps in Applied Steps.

Part A: Removing Duplicates

Power Query's Remove Duplicates option keeps the first occurrence of any combination of values you specify and removes all later duplicates. You choose which columns define uniqueness — in this case, Station Name and Period together.

WORD TO KNOW**Remove
Duplicates
(Power
Query)**

An option under Home → Remove Rows → Remove Duplicates. It keeps the first row for each unique combination of values in the columns you have selected, and removes all subsequent rows with the same combination.

To remove the duplicate: in the Power Query Editor, hold Ctrl and click the Station Name column header and the Period column header to select both. Then go to Home → Remove Rows → Remove Duplicates. Power Query keeps HC-007 (the first Victoria Park Q3 2024 row) and removes HC-041.

IMPORTANT NOTE

Remove Duplicates keeps the first occurrence of each combination. This means which row survives depends on the order of rows when the step runs. If you are not sure which row is more reliable, document your decision in the anomaly log before removing either one.

TRY IT

In the Power Query Editor, hold Ctrl and click the Station Name and Period column headers to select both. Go to Home → Remove Rows → Remove Duplicates. Check the Preview — you should now see only one row for Victoria Park Q3 2024.

Look at the Applied Steps panel. Rename the new step 'Remove Victoria Park Q3 duplicate'.

CHECK YOURSELF

After removing duplicates on Station Name and Period, how many rows should remain in the dataset?

Answer: 41 rows. The original dataset had 42 records (rows 2 to 43 in Excel). HC-041 was the duplicate. Removing it leaves 41 data records.

Part B: Handling Null Values

In Power Query, blank cells from Excel appear as null. You have three main choices for null values: replace them with a specific value, remove the rows that contain them, or leave them as null and flag them. The right choice depends on what the blank means.

WORD TO KNOW Null

Power Query's word for a blank or missing value. A null is not zero and not text — it represents the absence of any value. Most calculations in Power Query and Excel treat null differently from zero.

For HC-042 (Meadow Park), the operational data was never captured — the tracking system was not set up. The most honest treatment is to replace the nulls with a text marker like 'Not captured' in a new flag column, and leave the numeric columns as null so they are excluded from calculations rather than distorting them with a false zero.

To replace nulls in a specific column: right-click the column header in the Power Query Editor → Replace Values. In the Value to Find box, leave it empty (this matches null). In Replace With, type your marker text.

IMPORTANT NOTE

Replacing null with 0 in Total Hires would tell every formula that Meadow Park had zero hires — which is different from 'we don't know'. Zero would drag down the average, make the station appear to have performed badly, and be factually wrong. Leave numeric nulls as null unless you have a good reason to substitute a value.

TRY IT

In the Power Query Editor, right-click the Total Hires column header. Choose Replace Values. Leave the 'Value to Find' box empty. In 'Replace With', type Not captured. Click OK. Repeat for Avg Hire Duration, Total Revenue, and Customer Satisfaction Score.

Alternatively: add a Conditional Column called 'Data Completeness' — if Total Hires equals null then 'Incomplete record', otherwise 'Complete'. This flags HC-042 without modifying the numeric columns.

Rename each new Applied Step clearly: 'Flag HC-042 incomplete record' is more useful than 'Added Conditional Column1'.

CHECK YOURSELF

What is the difference between replacing a null with 0 and leaving it as null in the Total Hires column?

Answer: Replacing with 0 tells formulas that Meadow Park hired zero bikes — which is factually wrong and will drag down averages. Leaving it as null means AVERAGE and SUM skip the row entirely, which correctly excludes unrecorded data from calculations. Document whichever decision you make in the anomaly log.

Spot the Mistake

Marcus has cleaned the data. He did it by hand.

Marcus has sent you this message:

"I fixed all the data issues for you. I went into the Excel sheet and typed the correct values directly into the cells. I changed HC-030 Zone to South. I also deleted row 32 because those 8 hires looked wrong. Took me about 10 minutes."

"I did not use Power Query because it seemed complicated for only a few changes."

— Marcus

Marcus has made two significant mistakes. What are they?

TRY IT

Think about what Marcus did differently from the Power Query approach. Consider: what record exists of his changes? What happened to the deleted row? Can his work be reviewed, repeated, or undone?

ANSWER

Mistake 1 — No audit trail. Editing cells directly in Excel leaves no record. There is no way to see what value was there before, why it was changed, or who changed it. Brian cannot verify Marcus's work, and neither can an assessor.

Mistake 2 — Permanent deletion. Deleting row 32 (HC-031) removed it from the source data permanently. It is gone from totals, averages, and the record count — but there is no documentation of what was removed or why. If the 8-hire figure turns out to be correct (a genuine low quarter), the deletion has introduced a different error.

Power Query would have recorded both actions as named, reversible steps. The source data would be untouched. The step history would show exactly what changed and when.

Mini-Mission

A two-minute end-of-pack challenge

Priya has sent a message:

"Before we run the analysis, I need to know three things: which anomaly you fixed in Power Query and how, which records you flagged rather than fixed and why, and how I can see your audit trail if I need to check it."

— Priya

You have two minutes. Write three sentences — one per question.

TRY IT

Check your Applied Steps panel and your anomaly log. Then write your three-sentence answer.

ANSWER

1. HC-030 (South Station, Q2 2024) had Zone = 'West' — a clear entry error. This was corrected to 'South' using Replace Values in Power Query, and the step is recorded in the Applied Steps panel.
2. Five records (HC-010, HC-015, HC-019, HC-031, HC-035) were flagged using a conditional column rather than corrected, because the correct values are unknown — changing them without evidence would introduce new errors.
3. The audit trail is in the Power Query Editor: Data tab → Queries & Connections → double-click the Hire_Data query. The Applied Steps panel shows every action taken, in order, with names and the ability to click each step to review the data at that point.

You Can Now Answer

By the end of this pack, you should be able to answer each of these questions without looking anything up.

1. How do I open the Power Query Editor from an Excel table?
2. What does the Applied Steps panel show, and why is it useful?
3. How do I check and correct a column's data type in Power Query?
4. How do I fix a specific incorrect value in a column using Power Query?
5. How do I filter out rows in Power Query without deleting the source data?
6. How do I add a flag column that marks records for investigation?
7. How do I remove duplicate rows in Power Query by specifying which columns define uniqueness?
8. What is the difference between replacing a null with zero and leaving it as null?
9. How do I load the cleaned data back into Excel after I finish in Power Query?

If You Got Stuck

Four common stuck-points, and where to go to get unstuck.

"I cannot find From Table/Range on the Data tab." Go back to Chunk 1. Make sure you have clicked inside the Hire_Data table first — if no cell in the table is selected, the option may be greyed out. Also check you are looking in the Get & Transform Data section, not the Connections section.

"Power Query changed the type of a column I did not want to change." Go back to Chunk 2. This is usually the automatic 'Changed Type' step that Power Query adds when it first loads the data. Click that step in Applied Steps, review what it changed, and either delete it and re-add with correct types or edit its settings.

"Replace Values changed a Zone I did not want to change." Go back to Chunk 3. Replace Values acts on the whole column. If West is a valid zone for some stations, you need a different approach — filter to the affected rows first, or edit the source value directly. Delete the Replaced Value step from Applied Steps to undo.

"Close & Load created a new sheet but I cannot see my changes." Go back to Chunk 6. The cleaned data loads to a new sheet, not the original Hire_Data sheet. Look for a new tab in your workbook — it may be called something like 'Hire_Data (2)' or the name you gave the query. The original sheet is deliberately unchanged.